

O & Soergel, lec 4: Overview IV

1) Crash course on equivariant $\mathcal{D}$ -modules

1.1) Basics

$F \curvearrowright Y \leftarrow \text{smooth variety}$
 $\uparrow$
alg. group

Situations we care about:

1) $Y = G/B, F = N$ (or $Y = G, F = N \times B$)

2) $Y = G/B, F = B$ (or $Y = G, F = B \times B$)

$F \curvearrowright Y \rightsquigarrow$ bounded derived category $\mathcal{D}^b(Y/F) =$
 F -equivariant $\mathcal{D}$ -modules on Y .

• Has t -structure w. heart $\mathcal{D}_Y\text{-mod}^F \leftarrow$ abelian
cat. of F -equiv. $\mathcal{D}$ -modules.

• in some cases ($F = \{1\}$ or ex. 1)):

$$\mathcal{D}^b(\mathcal{D}_Y\text{-mod}^F) \xrightarrow{\sim} \mathcal{D}^b(Y/F)$$

• not equivalent: $Y = \text{pt}, H = G_m$.

1.2) Classification of irreducible objects.

(in $\mathcal{D}_Y\text{-mod}^F$). We'll need an assumption:

(*) F has fin. many orbits on Y : $Y = \bigsqcup_{i=1}^k Y_i$, $Y_i = F/F_i$

$F_i^\circ \subset F_i$ $\swarrow$ connected component of 1
 $\downarrow$

F_i/F_i° is finite group.

Fact: Under (*), $\text{Irr}(\mathcal{D}_Y\text{-mod}^F) \ni \text{IC}(Y_i, \mathcal{G}_i)$

$$\begin{array}{ccc} & \downarrow \wr & \uparrow \\ \bigsqcup_{i=1}^k \text{Irrep}(F_i/F_i^\circ) & & \mathcal{G}_i \end{array}$$

Examples: 1) If $\bar{Y}_i = Y$, $\mathcal{G}_i = \text{triv} \Rightarrow \text{IC}(Y_i, \text{triv}) = \mathcal{D}_Y$.

2) $Y = G/B$, $F = N$ or B , satisfies (*) b/c

Bruhat decomposition $G = \bigsqcup_{w \in W} N \dot{w} B$

lift of $w \in W = N_G(H)/H$ to $N_G(H)$

N - or B -orbits in $G/B \xleftrightarrow{\sim} W$

Stabilizers for N -action: $N_w = \{n \in N \mid n \dot{w} B = \dot{w} B\}$

is unipotent hence connected.

$$\text{Fact: } \text{Irr}(\mathcal{D}_{G/B}\text{-mod}^N) \xrightarrow{\sim} W$$

$$\text{Agrees w. } \mathcal{D}_{G/B}\text{-mod}^N \xrightarrow{\sim} \text{OP}^r \text{ (Lec 3)}$$

$$F=B: \text{ stabilizers are } H \ltimes N_w \text{ so } \text{Irr}(\mathcal{D}_{G/B}\text{-mod}^B) \xrightarrow{\sim} W$$

1.3) Construct irreducible objects

$\mathcal{D}_Y\text{-mod}^F \supset$ full subcategory of "holonomic modules."

- $\mathcal{O}_Y$ is holonomic.

- If (*) holds: holonomic $\Leftrightarrow$ finite length.

$\mathcal{D}_{\text{hol}}^6(Y/F) = \{\text{complexes w. holonomic homology}\}$
 $\uparrow$
has "6-functor formalism."

e.g. F -equiv. map $f: Y_1 \rightarrow Y_2 \leadsto$

1) Pullback functor $f^*: \mathcal{D}_{\text{hol}}^6(Y_2) \rightarrow \mathcal{D}_{\text{hol}}^6(Y_1)$

$$(fg)^* = g^* f^*, \quad f^* \mathcal{O}_{Y_1} \simeq \mathcal{O}_{Y_2}$$

2) Pushforward functor f_* , right adjoint.

Assume (*): inclusion $\iota_i: Y_i \hookrightarrow Y_i$ (loc. closed)

a) $\iota_{i,*} \mathcal{O}_{Y_i} \in \mathcal{D}_{\text{hol}}^6(Y/F)$ is cohom. deg ≥ 0 .

$$b) \text{IC}(\mathcal{Y}_i, \text{triv}) \hookrightarrow \mathcal{H}^0(\iota_{i,*} \mathcal{O}_{\mathcal{Y}_i}) \quad (1)$$

supported (as q -coh sheaf) on $\overline{\mathcal{Y}}_i$ & coker (1) & $\mathcal{H}^j(\iota_{i,*} \mathcal{O}_{\mathcal{Y}_i})$ are supported on $\overline{\mathcal{Y}}_i \setminus \mathcal{Y}_i$.

$$c) \text{ If } \overline{\mathcal{Y}}_i \text{ is smooth, } \overline{\iota}_i: \overline{\mathcal{Y}}_i \hookrightarrow \mathcal{Y} \Rightarrow \\ \text{IC}(\mathcal{Y}_i, \text{triv}) = \overline{\iota}_{i,*} \mathcal{O}_{\overline{\mathcal{Y}}_i}.$$

Decomposition thm (BBDG):

$$F\text{-equiv. projective } \varphi: \underset{\substack{\uparrow \\ \text{smooth}}}{\widetilde{\mathcal{Y}}_i} \rightarrow \overline{\mathcal{Y}}_i$$

$$\text{Then } \varphi_* \mathcal{O}_{\widetilde{\mathcal{Y}}_i} \simeq \bigoplus_{\substack{\mathcal{Y}_j \subset \overline{\mathcal{Y}}_i, \\ \mathcal{G}_j}} \text{IC}(\mathcal{Y}_j, \mathcal{G}_j)[?]^{\oplus ?}$$

If φ is iso over $\mathcal{Y}_i \Rightarrow$ only term w. $\mathcal{Y}_j = \mathcal{Y}_i$ is $\text{IC}(\mathcal{Y}_i, \text{triv})$ (mult. is 1 & cohom. deg is 0).

1.3) Equivariant cohomology

$$(\check{\text{Ext}}_{\mathcal{D}^b(\mathcal{Y}/F)}^*(\mathcal{O}_{\mathcal{Y}}, \bullet))$$

Base field is $\mathbb{C}$

X is topol. space w. topological $F \curvearrowright X$

→ equivariant cohomology $H_F^*(X) = H^*((E \times X)/F)$, where

- E is contractible top. space w. $F \curvearrowright E$

- & $\exists$ topological princ. F -bundle $E \rightarrow E/F$.

If $\exists$ princ. bundle $X \rightarrow X/F \Rightarrow (E \times X)/F \rightarrow X/F$
is loc. triv. bundle w. fiber E (contractible) $\Rightarrow$

$$H_F^*(X) \xleftarrow{\sim} H^*(X/F)$$

in general, this is correct cohom. of quotient $F \curvearrowright X$

$H_F^*(X)$ is $H_F^*(pt)$ -algebra.

Spec. cases:

1) F is unipotent $\Rightarrow F \xrightarrow{\sim} \mathbb{A}^{\dim F}$ is contractible

Take $E = F \Rightarrow H_F^*(X) = H^*(X)$.

2) $F = \mathbb{C}_m$, $E = \mathbb{C}^\infty \setminus \{0\}$

$\{(z_i)_{i \geq 0} \mid \text{only fin. many } z_i \neq 0\}$

$E/F = \mathbb{CP}^\infty \Rightarrow H_F^*(pt) = H^*(\mathbb{CP}^\infty) = \mathbb{C}[x]$, $x \in H^2(\mathbb{CP}^\infty)$.

More generally, $F = H \Rightarrow H_H^*(pt) \xrightarrow{\sim} \mathbb{C}[h]$.

3) $H_B^*(X) \xrightarrow{\sim} H_H^*(X)$ (exercise).

Remarks: a) G is conn. reductive $\Rightarrow H_G^*(pt) = \mathbb{C}[\sigma]^{G^*}$ deg 2
 $\Rightarrow H_F^{\text{odd}}(pt) = \{0\} \Rightarrow H_F^*(pt)$ is usual commut. alg.

b) $(E \times X)/F \rightarrow E/F$ fiber bundle w. fiber $X \leadsto$

$$H_F^*(X) \otimes_{H_F^*(pt)} \mathbb{C}_0 \rightarrow H^*(X) \quad (2)$$

If $H^{\text{odd}}(X) = \{0\}$, this is iso.

1.4) Case of flag & Bott-Samelson varieties.

I) $H_B^*(G/B) = H_B^*(G)$ is $H_B^*(pt) = \mathbb{C}[\lambda]$ -algebra
 $\mathbb{C}[\lambda]^{\text{cow}} = \mathbb{C}[\lambda] / \mathbb{C}[\lambda] \mathbb{C}[\lambda]_+^w$

II) $H_B^*(G/B) = H_{B \times B}^*(G)$ is $\mathbb{C}[\lambda] \otimes \mathbb{C}[\lambda]$ -algebra
 $\mathbb{C}[\lambda] \otimes \mathbb{C}[\lambda]^w \mathbb{C}[\lambda]$.

III) $S = \{\text{simple refl.'ns in } w\} \ni s \leadsto \text{minimal parabolic } P_s \supset B$
 $\underline{w} = (s_1, \dots, s_k) \in S^k \leadsto \text{Bott-Samelson variety } BS_{\underline{w}}$

$B^k \curvearrowright P_{s_1} \times P_{s_2} \times \dots \times P_{s_k} : (b_1, \dots, b_k) \cdot (p_1, \dots, p_k) = (p_1 b_1^{-1}, b_1 p_2 b_2^{-1}, \dots, b_{k-1} p_k b_k^{-1})$, $BS_{\underline{w}} = P_{s_1} \times \dots \times P_{s_k} / B^k$ (dim = k)

$[p_1, \dots, p_k] := \text{orbit of } (p_1, \dots, p_k)$

Two maps:

$$BS_{(s_1, \dots, s_k)} \rightarrow BS_{(s_1, \dots, s_{k-1})}, [p_1, \dots, p_k] \mapsto [p_1, \dots, p_{k-1}] \quad (3)$$

$$\rightarrow G/B \quad \nwarrow \quad p_1 \dots p_k B \quad (4)$$

(4) $\leadsto H_B^*(BS_{\underline{w}})$ is graded $H_B^*(G/B)$ -algebra

(3) is $\mathbb{P}^1$ -bundle $\leadsto H_B^*(BS_{\underline{w}}) = BS_{\underline{w}} \langle -k \rangle$
 $\uparrow$
 BS bimodule (Lec 2)

$$H^*(BS_{\underline{w}}) = H_B^*(BS_{\underline{w}}) \otimes_{\mathbb{C}[y]} \mathbb{C}_0 = BS_{\underline{w}} \otimes_{\mathbb{C}[y]} \mathbb{C}_0 \langle -k \rangle.$$

Remark: $w \in W$ w. reduced expression $\underline{w} = (s_1, \dots, s_\ell)$

(4): $\varphi_{\underline{w}}: BS_{\underline{w}} \rightarrow G/B$ has image $\overline{BwB/B}$ & is resolution of singularities onto its image.

1.5) $\text{Ext}_{\mathcal{D}^b(Y/F)}^*(\mathcal{O}_Y, ?)$

Fact: Y is smooth $\Rightarrow \text{Ext}_{\mathcal{D}^b(Y/F)}^*(\mathcal{O}_Y, \mathcal{O}_Y) \simeq H_F^*(Y)$.

Q: Under (*) from Sec 1.2, how to compute

$$\text{Ext}_{\mathcal{D}^b(Y/F)}^*(\mathcal{O}_Y, \text{IC}(Y_i, \text{triv}))?$$

Easy case: $\overline{Y}_i$ is smooth: $\text{Ext}^*(\mathcal{O}_Y, \overline{\tau}_{i,*} \mathcal{O}_{\overline{Y}_i})$

$\Rightarrow \text{Ext}^*(\bar{\mathcal{L}}_i^* \mathcal{O}_Y, \mathcal{O}_{\bar{Y}_i}) = \text{Ext}^*(\mathcal{O}_{\tilde{Y}_i}, \mathcal{O}_{\bar{Y}_i}) = H_F^*(\bar{Y}_i)$ as graded $H_F^*(Y)$ -module.

General case: Define equivariant intersection cohomology $IH_F^*(\bar{Y}_i) = \text{Ext}_{\mathcal{D}^b(Y/F)}^*(\mathcal{O}_Y, IC(Y_i, \text{triv})[\dim Y_i])$.

Cov. of Decomp. thm: If $(*)$ holds & $\tilde{Y}_i \rightarrow \bar{Y}_i$ is F -equiv. resol'n of sing's $\Rightarrow$ graded $H_F^*(Y)$ -module $IH_F^*(\bar{Y}_i) \langle \dim Y_i \rangle \simeq$

$$IH_F^*(\bar{Y}_i) \oplus \bigoplus_{j | \bar{Y}_j \neq Y_i, \sigma_j} \text{Ext}^*(\mathcal{O}_Y, IC(Y_j, \sigma_j)[?])^{\oplus ?}$$

Example: $F=B, Y=G/B \Rightarrow H_B^*(G/B) = \mathbb{C}[Y] \otimes_{\mathbb{C}[Y]^W} \mathbb{C}[Y]$

$Y_i = BwB/B, \dim Y_i = \ell(w), \tilde{Y}_i = BS_{\underline{w}}, \varphi = \varphi_{\underline{w}}$

$\mathcal{B}'_{\underline{w}} := IH_B^*(\bar{Y}_i) \Rightarrow$

$$BS_{\underline{w}} \simeq \mathcal{B}'_{\underline{w}} \oplus \bigoplus_{u | \ell(u) < \ell(w)} \mathcal{B}'_u \langle ? \rangle^{\oplus ?}$$

We'll mention: $\mathcal{B}'_{\underline{w}}$ is indecomposable as bimodule

$$\Rightarrow \mathcal{B}_{\underline{w}} \simeq \mathcal{B}'_{\underline{w}}$$

↑
indec. Soergel bimodule.